

MATH I - Week 1

Fall 2026

Practical sessions material — Instructor solutions ¶

The original exercises are followed immediately by worked solutions. Equivalent correct methods should also be accepted.

Exercise 1: Decimal representations

Classify each decimal as terminating, repeating or nonrepeating. State whether the corresponding number is rational or irrational.

1. 0.375
2. $0.272727 \dots$
3. 1.101001000100001 ...
4. 3.14159265 ...
5. $0.123123123 \dots$

Solution 1

1. 0.375 is **terminating**, so it is rational:

$$0.375 = \frac{375}{1000} = \frac{3}{8}.$$

2. $0.272727 \dots = 0.\overline{27}$ is **repeating**, so it is rational:

$$0.\overline{27} = \frac{27}{99} = \frac{3}{11}.$$

3. 1.101001000100001 ... is nonterminating and **nonrepeating**: the number of zeros between successive 1s keeps increasing. It is therefore irrational.

4. As written, 3.14159265 ... is **not sufficient to determine the classification**. Finitely many displayed digits followed by an ellipsis do not specify the remaining digits. If the intended number is

$$\pi = 3.14159265 \dots,$$

then it is nonterminating, nonrepeating and irrational.

5. $0.123123123 \dots = 0.\overline{123}$ is **repeating**, so it is rational:

$$0.\overline{123} = \frac{123}{999} = \frac{41}{333}.$$

Principle: Every terminating or eventually repeating decimal is rational. A nonterminating decimal that never becomes periodic is irrational.

Exercise 2: Intervals

Write each set using interval notation.

1. $\{x \in \mathbb{R} : -3 \leq x < 4\}$
2. $\{x \in \mathbb{R} : x \neq 3\}$

Then represent each set on the real number line.

Solution 2

1. Since -3 is included and 4 is excluded,

$$[-3, 4).$$

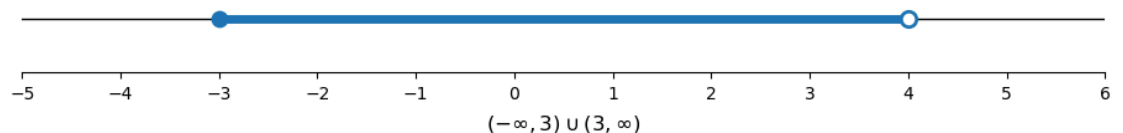
2. All real numbers except 3 are included:

$$(-\infty, 3) \cup (3, \infty).$$

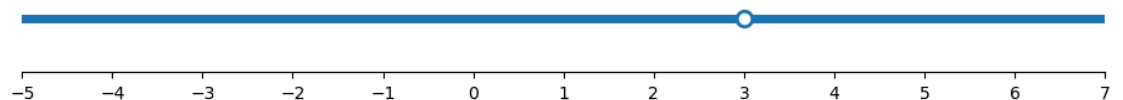
On a number line, use a **closed dot** at an included finite endpoint and an **open dot** at an excluded endpoint. Infinity is never an included endpoint, so it always takes a parenthesis.

```
In [1]: 1 import matplotlib.pyplot as plt
2
3 fig, axes = plt.subplots(2, 1, figsize=(9, 2.8), constrained_layout=True)
4
5 # [-3, 4)
6 ax = axes[0]
7 ax.axhline(0, color='black', linewidth=1)
8 ax.plot([-3, 4], [0, 0], color='tab:blue', linewidth=5)
9 ax.scatter([-3], [0], s=80, color='tab:blue', zorder=3)
10 ax.scatter([4], [0], s=80, facecolors='white', edgecolors='tab:blue',
11 ax.set_title(r'$[-3,4)$')
12 ax.set_xlim(-5, 6); ax.set_ylim(-0.5, 0.5); ax.set_yticks([]); ax.set
13
14 # R minus {3}
15 ax = axes[1]
16 ax.axhline(0, color='tab:blue', linewidth=5)
17 ax.scatter([3], [0], s=80, facecolors='white', edgecolors='tab:blue',
18 ax.set_title(r'$(-\infty,3)\cup(3,\infty)$')
19 ax.set_xlim(-5, 7); ax.set_ylim(-0.5, 0.5); ax.set_yticks([]); ax.set
20
21 for ax in axes:
22     for side in ['left', 'right', 'top']:
23         ax.spines[side].set_visible(False)
24 plt.show()
25
```

$[-3, 4)$



$(-\infty, 3) \cup (3, \infty)$



Exercise 3: From intervals to inequalities

Write each interval using inequalities.

- a. $[-2, 5)$, b. $(-\infty, 4]$, c. $(1, \infty)$.

Solution 3

$$\text{a. } [-2, 5) \iff \boxed{-2 \leq x < 5},$$

$$\text{b. } (-\infty, 4] \iff \boxed{x \leq 4},$$

$$\text{c. } (1, \infty) \iff \boxed{x > 1}.$$

A square bracket at a finite endpoint means equality is allowed; a parenthesis means that endpoint is excluded.

About irrational numbers

The set of irrational numbers is

$$\mathbb{R} \setminus \mathbb{Q}.$$

An irrational number cannot be expressed as

$$\frac{p}{q}, \quad p, q \in \mathbb{Z}, \quad q \neq 0.$$

Exercise 4: Rational or irrational?

Classify each number as rational or irrational. Simplify before classifying.

$$\text{a. } \sqrt{3}, \quad \text{b. } \sqrt{25}, \quad \text{c. } \sqrt{50},$$

$$\text{i. } (1 + \sqrt{5})(1 - \sqrt{5}), \quad \text{j. } \frac{1}{\sqrt{3}}.$$

Solution 4

a. $\sqrt{3}$ is irrational because 3 is not a perfect square.

b.

$$\sqrt{25} = 5,$$

so it is rational (and also an integer and a natural number).

c.

$$\sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2},$$

so it is irrational.

i. Using the difference of two squares,

$$(1 + \sqrt{5})(1 - \sqrt{5}) = 1 - (\sqrt{5})^2 = 1 - 5 = -4,$$

so it is rational (and an integer).

j. Rationalizing the denominator, the result is

$$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3},$$

which is apparently an irrational number.

Complex numbers

Exercise 5: Basic complex arithmetic

Given

$$z_1 = 3 + 2i, \quad z_2 = 1 - 4i,$$

calculate

$$z_1 + z_2, \quad z_1 - z_2, \quad z_1 z_2, \quad \frac{z_1}{z_2}.$$

Write every answer in the form $a + bi$.

Solution 5

$$z_1 + z_2 = (3 + 2i) + (1 - 4i) = \boxed{4 - 2i}.$$

$$z_1 - z_2 = (3 + 2i) - (1 - 4i) = \boxed{2 + 6i}.$$

Using $i^2 = -1$,

$$z_1 z_2 = (3 + 2i)(1 - 4i) = 3 - 12i + 2i - 8i^2 = \boxed{11 - 10i}.$$

For division, multiply numerator and denominator by the conjugate $1 + 4i$:

$$\frac{3 + 2i}{1 - 4i} = \frac{(3 + 2i)(1 + 4i)}{(1 - 4i)(1 + 4i)} = \frac{-5 + 14i}{17} = \boxed{-\frac{5}{17} + \frac{14}{17}i}.$$

Exercise 6: Complex conjugate and modulus

For each number, find its conjugate and modulus:

$$3 + 4i, \quad -2 + 5i, \quad 6i, \quad -7.$$

Optionally verify that $z\bar{z} = |z|^2$.

Solution 6

For $z = a + bi$,

$$\bar{z} = a - bi, \quad |z| = \sqrt{a^2 + b^2}.$$

z	$\bar{z}$	$ z $	Verification
$3 + 4i$	$3 - 4i$	5	$(3 + 4i)(3 - 4i)$ $= 25$ $= 5^2$

z	$\bar{z}$	$ z $	Verification
$-2 + 5i$	$-2 - 5i$	$\sqrt{29}$	$\begin{aligned} &(-2 + 5i)(-2 - 5i) \\ &= 29 \end{aligned}$
$6i$	$-6i$	6	$\begin{aligned} &(6i)(-6i) \\ &= 36 \end{aligned}$
-7	-7	7	$\begin{aligned} &(-7)(-7) \\ &= 49 \end{aligned}$

Exercise 7: Complex division

Express in algebraic form $a + bi$:

$$\frac{1}{1+i}.$$

Solution 7

Multiply by the conjugate of the denominator:

$$\frac{1}{1+i} \cdot \frac{1-i}{1-i} = \frac{1-i}{(1+i)(1-i)} = \frac{1-i}{2} = \boxed{\frac{1}{2} - \frac{1}{2}i}.$$

Exercise 8: Quadratic equations

Solve over $\mathbb{C}$:

$$x^2 + 9 = 0.$$

Solution 8

$$x^2 + 9 = 0 \iff x^2 = -9.$$

Since $i^2 = -1$,

$$x = \boxed{\pm 3i}.$$

Check: $(3i)^2 + 9 = -9 + 9 = 0$, and the same holds for $-3i$.

Exercise 9: Complex plane

Represent $2 + 3i$ in the complex plane. Determine its real part, imaginary part, modulus and quadrant.

What changes for $2 - 3i$?

Solution 9

For $z = 2 + 3i$,

$$\operatorname{Re}(z) = 2, \quad \operatorname{Im}(z) = 3,$$

and the corresponding point is $(2, 3)$. Its modulus is

$$|z| = \sqrt{2^2 + 3^2} = \boxed{\sqrt{13}}.$$

Because both coordinates are positive, it lies in **Quadrant I**.

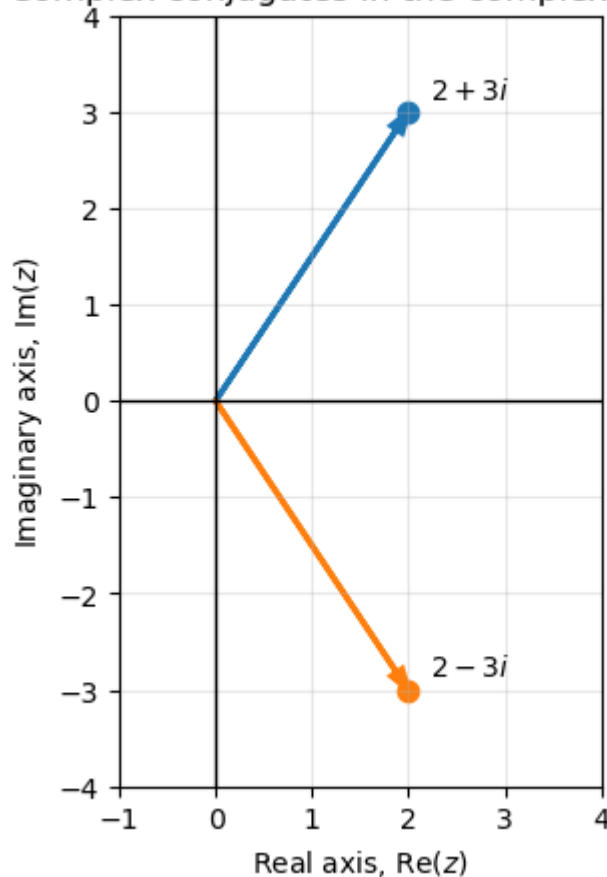
For $w = 2 - 3i$, the point is $(2, -3)$:

$$\operatorname{Re}(w) = 2, \quad \operatorname{Im}(w) = -3, \quad |w| = \sqrt{13}.$$

It lies in **Quadrant IV**. The two numbers are conjugates and their points are reflections across the real axis.

```
In [5]: 1 import matplotlib.pyplot as plt
2
3 fig, ax = plt.subplots(figsize=(6, 5))
4 ax.axhline(0, color='black', linewidth=1)
5 ax.axvline(0, color='black', linewidth=1)
6 ax.grid(True, alpha=0.3)
7
8 for x, y, label, color in [(2, 3, r'$2+3i$', 'tab:blue'), (2, -3, r'$2-3i$', 'tab:orange')]:
9     ax.arrow(0, 0, x, y, length_includes_head=True, head_width=0.14,
10             head_length=0.22, color=color, linewidth=2)
11     ax.scatter(x, y, color=color, s=55)
12     ax.annotate(label, (x, y), xytext=(8, 5), textcoords='offset point',
13                color=color, size=12)
14
15 ax.set_xlim(-1, 4); ax.set_ylim(-4, 4)
16 ax.set_aspect('equal', adjustable='box')
17 ax.set_xlabel(r'Real axis, $\operatorname{Re}(z)$')
18 ax.set_ylabel(r'Imaginary axis, $\operatorname{Im}(z)$')
19 ax.set_title('Complex conjugates in the complex plane')
20 plt.show()
```

Complex conjugates in the complex plane



Exercise 10: Polar form

Write $z = 1 + i$ in trigonometric and exponential forms. Give θ in radians.

Solution 10

The modulus is

$$r = |z| = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Since $(1, 1)$ lies in Quadrant I,

$$\theta = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}.$$

Therefore, the trigonometric form is

$$z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right),$$

and the exponential form is

$$z = \sqrt{2} e^{i\pi/4}.$$

Exercise 11: de Moivre's formula

Use de Moivre's formula to calculate

$$(\sqrt{3} + i)^4.$$

Check the result by algebraic multiplication.

Solution 11

First write $\sqrt{3} + i$ in polar form. Its modulus and argument are

$$r = \sqrt{(\sqrt{3})^2 + 1^2} = 2, \quad \theta = \frac{\pi}{6}.$$

Thus,

$$\sqrt{3} + i = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right).$$

By de Moivre's formula,

$$(\sqrt{3} + i)^4 = 2^4 \left(\cos \frac{4\pi}{6} + i \sin \frac{4\pi}{6} \right) = 16 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right).$$

Therefore,

$$(\sqrt{3} + i)^4 = 16 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) = \boxed{-8 + 8\sqrt{3}i}.$$

Algebraic check:

$$(\sqrt{3} + i)^2 = 3 + 2\sqrt{3}i + i^2 = 2 + 2\sqrt{3}i,$$

and

$$(2 + 2\sqrt{3}i)^2 = 4 + 8\sqrt{3}i - 12 = -8 + 8\sqrt{3}i.$$

Exercise 12: Roots of a complex number

Solve

$$z^3 = -8$$

and represent the solutions in the complex plane.

Solution 12

Write -8 in polar form:

$$-8 = 8e^{i(\pi+2k\pi)}.$$

The cube roots have modulus

$$\sqrt[3]{8} = 2$$

and arguments

$$\theta_k = \frac{\pi + 2k\pi}{3}, \quad k = 0, 1, 2.$$

Hence the three arguments are

$$\frac{\pi}{3}, \quad \pi, \quad \frac{5\pi}{3}.$$

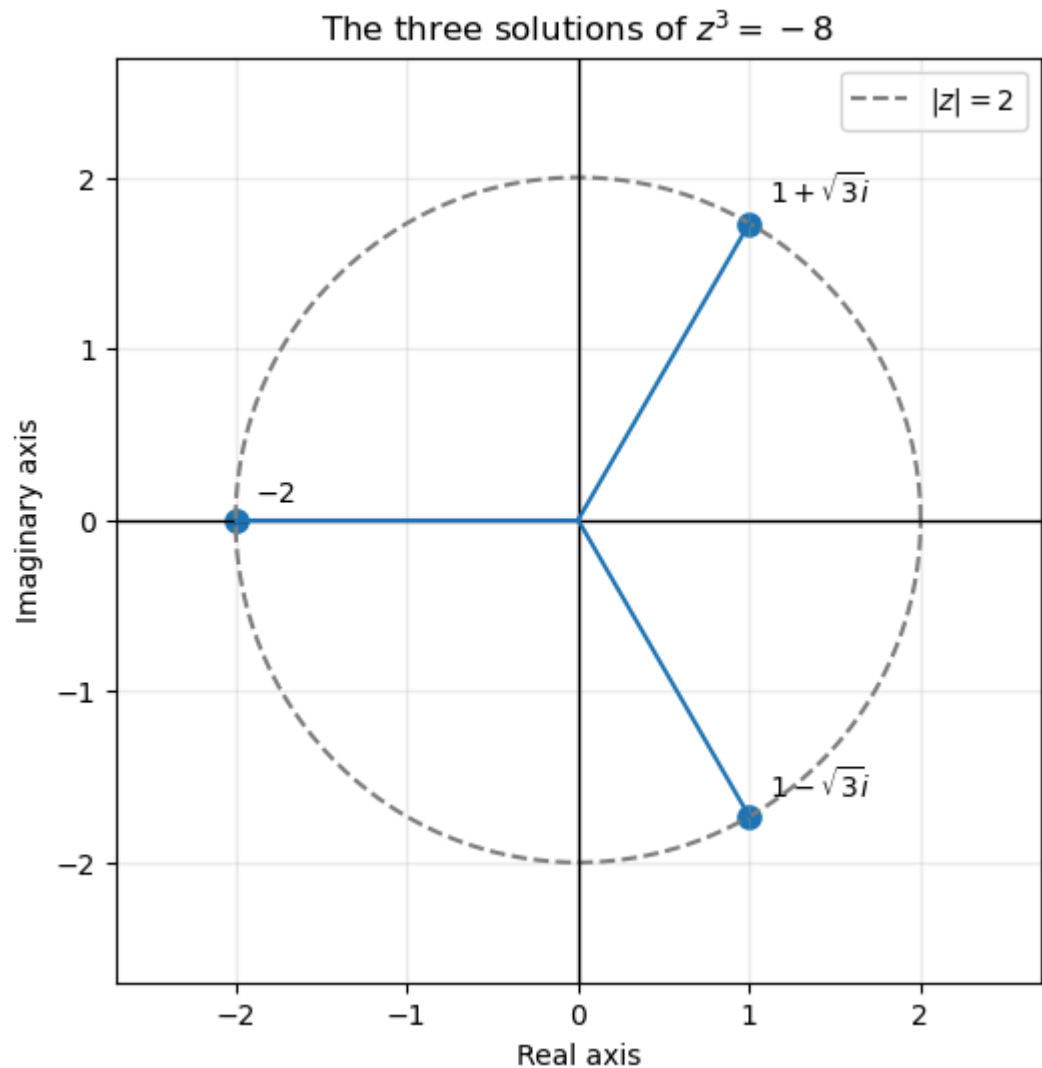
The roots are therefore

$$\begin{aligned} z_0 &= 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = \boxed{1 + \sqrt{3}i}, \\ z_1 &= 2(\cos \pi + i \sin \pi) = \boxed{-2}, \\ z_2 &= 2 \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right) = \boxed{1 - \sqrt{3}i}. \end{aligned}$$

The roots lie on the circle $|z| = 2$ and are separated by angles of $2\pi/3$.

In [8]:

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 roots = np.array([1 + np.sqrt(3)*1j, -2 + 0j, 1 - np.sqrt(3)*1j])
5
6 fig, ax = plt.subplots(figsize=(6, 6))
7 theta = np.linspace(0, 2*np.pi, 400)
8 ax.plot(2*np.cos(theta), 2*np.sin(theta), linestyle='--', color='gray')
9 ax.axhline(0, color='black', linewidth=1)
10 ax.axvline(0, color='black', linewidth=1)
11
12 labels = [r'$1+\sqrt{3}i$', r'$-2$', r'$1-\sqrt{3}i$']
13 for z, label in zip(roots, labels):
14     ax.scatter(z.real, z.imag, s=65, color='tab:blue')
15     ax.plot([0, z.real], [0, z.imag], color='tab:blue', linewidth=1.5)
16     ax.annotate(label, (z.real, z.imag), xytext=(7, 7), textcoords='ot
17
18 ax.set_aspect('equal', adjustable='box')
19 ax.set_xlim(-2.7, 2.7); ax.set_ylim(-2.7, 2.7)
20 ax.set_xlabel('Real axis')
21 ax.set_ylabel('Imaginary axis')
22 ax.set_title(r'The three solutions of $z^3=-8$')
23 ax.grid(True, alpha=0.25)
24 ax.legend()
25 plt.show()
26
```



Final observations

- Simplify an expression before deciding whether it is rational or irrational.
- In interval notation, brackets indicate inclusion and parentheses indicate exclusion.
- For $z = a + bi$, the point in the complex plane is (a, b) and $|z| = \sqrt{a^2 + b^2}$.
- Polar form is particularly useful for powers and roots of complex numbers.